



Receipt & Invoice Digitizer

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Problem Statement Overview

Businesses and individuals handle numerous paper receipts and invoices, which are prone to loss, errors, and manual entry delays.

We built automated system for scanning, extracting and digitizing information from receipts and invoices using OCR.

Core Components

Frontend: Uses Streamlit to create a web-based dashboard with tabs for uploading receipts and viewing analytics.

API: Leverages Google Gemini Flash, a fast AI model that reads images and extracts itemised details into JSON.

Database: A SQLite database acts as a persistent vault, storing merchant names, dates, and costs indefinitely.

Database Structure

Uses sqlite3 to create a table called receipts.

It stores:

- Merchant Name & Date of Purchase
- Total Cost & Currency Type
- Raw JSON: A detailed list of every item bought.

This allows the app to remember your spending even after you close the browser

Image Preprocessing

We used following to "clean" the receipt:

Grayscale: Removes unnecessary colors.

Denoising: Clears graininess.

Thresholding: Makes text black and white.

This results in much higher extraction success rates.

AI Extraction: Gemini API

The core intelligence is the `analyze_receipt()` function.

It sends the cleaned image to Google's AI with a specific prompt.

The AI is instructed to "Extract receipt details into JSON."

By specifying a format, we ensure the computer can read the AI's response without errors

Instructing the AI

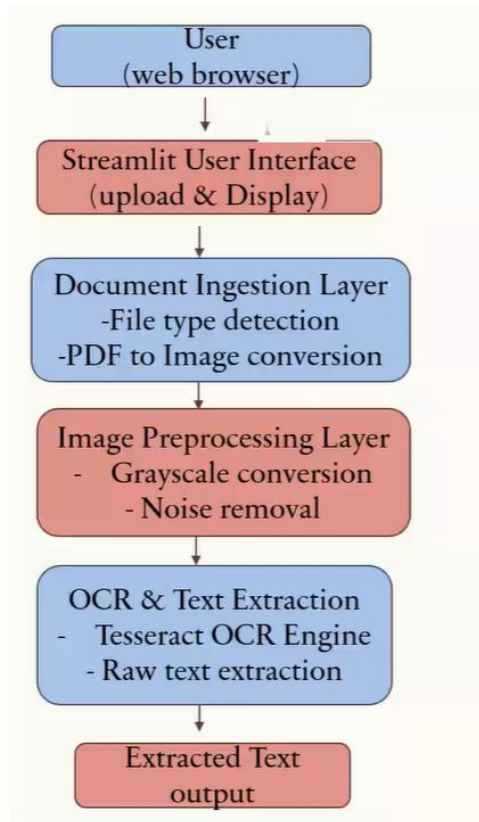
We use a structured text prompt to tell the AI exactly what we want. This prevents the AI from "chatting" and forces it to just give us the data. Strict Output: The prompt ends with "Return ONLY JSON."

JSON is like a digital checklist. It defines labels like "merchant" and "total". This makes it easy for our Python code to save the data directly into our SQLite database.

User Interface

- Multi-Format Upload: Supports JPG, PNG, and even multi-page PDFs.
- Side-by-Side View: Users see the original image vs the "cleaned" AI version.
- History Vault: A persistent table that lets users select a specific receipt to see its individual line items.
- Analytics dashboard: Includes filters, charts and graph.

System Architecture Diagram



Implemented Modules

Milestone 1:

- Document Ingestion & OCR
- Implement file upload.
- Preprocess images for OCR.
- Extract raw text from sample receipts.